

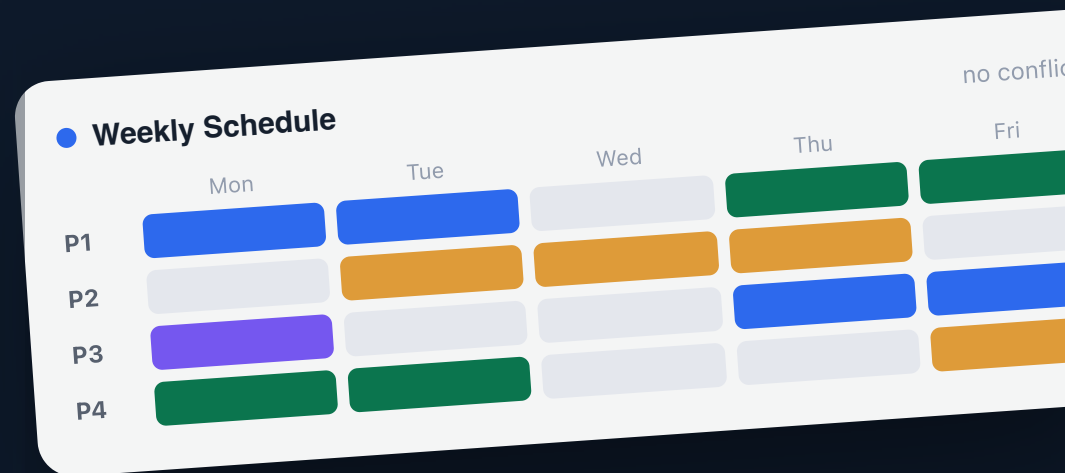
SAMPLE LESSON · CASE STUDY: AI SMART SCHEDULING

Can an Algorithm Schedule Better Than a Human?

A look inside a World Science Academy research session — where a messy real-world problem becomes award-winning research.

 LECTURER · EMILY CHOU

Optimization · Systems · Applied AI · 01



TODAY'S CHALLENGE

A team juggles people, time, and dozens of tasks.

**One delay throws everything off.
Can software schedule it better
than a human with a
spreadsheet?**

BY THE END, STUDENTS CAN

- ✓ Model a scheduling problem as variables and constraints
- ✓ Compare a hand-made schedule vs. an algorithm on real metrics
- ✓ Use AI tools to build a working prototype of the solution

WHY THIS IS HARD

Scheduling explodes fast

01 Combinatorial explosion

Just 10 tasks can be ordered 3,628,800 ways (10!). Checking them all by hand is hopeless.

02 Coupled constraints

Move one task and dozens of others shift — a spreadsheet can't keep up.

03 Conflicts surface too late

Clashes only appear after tasks are assigned — never before.

● **Manual Spreadsheet** 2 conflicts

	Mon	Tue	Wed	Thu	Fri
P1			!		
P2			!		
P3					
P4					

Two people booked for the same Wednesday slot — and lots of idle gaps nobody planned for.

Four steps from chaos to a plan

1 • Model

Turn the problem into variables and constraints.

2 • Search

Try many possible arrangements — far more than a human could.

3 • Score

Rate each one: fewer clashes, less idle time, deadlines met.

4 • Recommend

Return the best plan — and auto-resolve conflicts when things change.

The computer doesn't get tired or lose track — it can weigh thousands of options in a second.

KEY CONCEPTS

The language students build

Constraint

A rule the schedule must obey — no double-booking, respect time-off.

Objective function

The number you try to make best — fewest clashes, least idle time.

Optimization

Searching for the arrangement that scores the best.

Heuristic

A smart shortcut that finds a good answer fast when checking every option is impossible.

● Optimized Schedule						0 conflicts
	M	T	W	T	F	
P1						
P2						
P3						
P4						

Students do, not just listen

In-Person

- Race the algorithm: schedule by hand, then compare
- Think-Pair-Share on constraints & trade-offs
- Measure a schedule with real metrics
- Case-study discussion of a real system

Online

- Annotate a real scheduling case study
- Micro-module self-checks
- Small-group problem-solving pods
- Hands-on prototyping with AI tools

Which scheduling strategy wins?

Students measure four ways to build the same schedule — and weigh the trade-offs.

Strategy	Conflicts	Planning time	Idle time	Adapts
Manual (spreadsheet)	Many	Hours	High	Poor
Greedy algorithm	Few	Seconds	Medium	OK
Constraint solver	0	Seconds	Low	Good
AI-assisted	Few	Instant	Low	Excellent

Illustrative teaching figures — the point is the trade-off pattern, not exact values.

THE BIG IDEA

No schedule is perfect — you optimize for what matters

Faster, fairer, cheaper — you usually can't max all three. And more automation isn't always better: in Emily's real project, a KPI dashboard was **deliberately left out** because it would feel like surveillance and break the team's trust. **Good systems serve people, not just numbers.**

Optimize the right metric

Automation needs human trust

Edge cases decide real success

From tangled spreadsheets to one predictive board

A video-production team ran on brittle, hand-built Excel. A single delay meant hours of manual recalculation. The **AI Smart Scheduling System** collapsed it all into one dynamic board that auto-resolves conflicts.

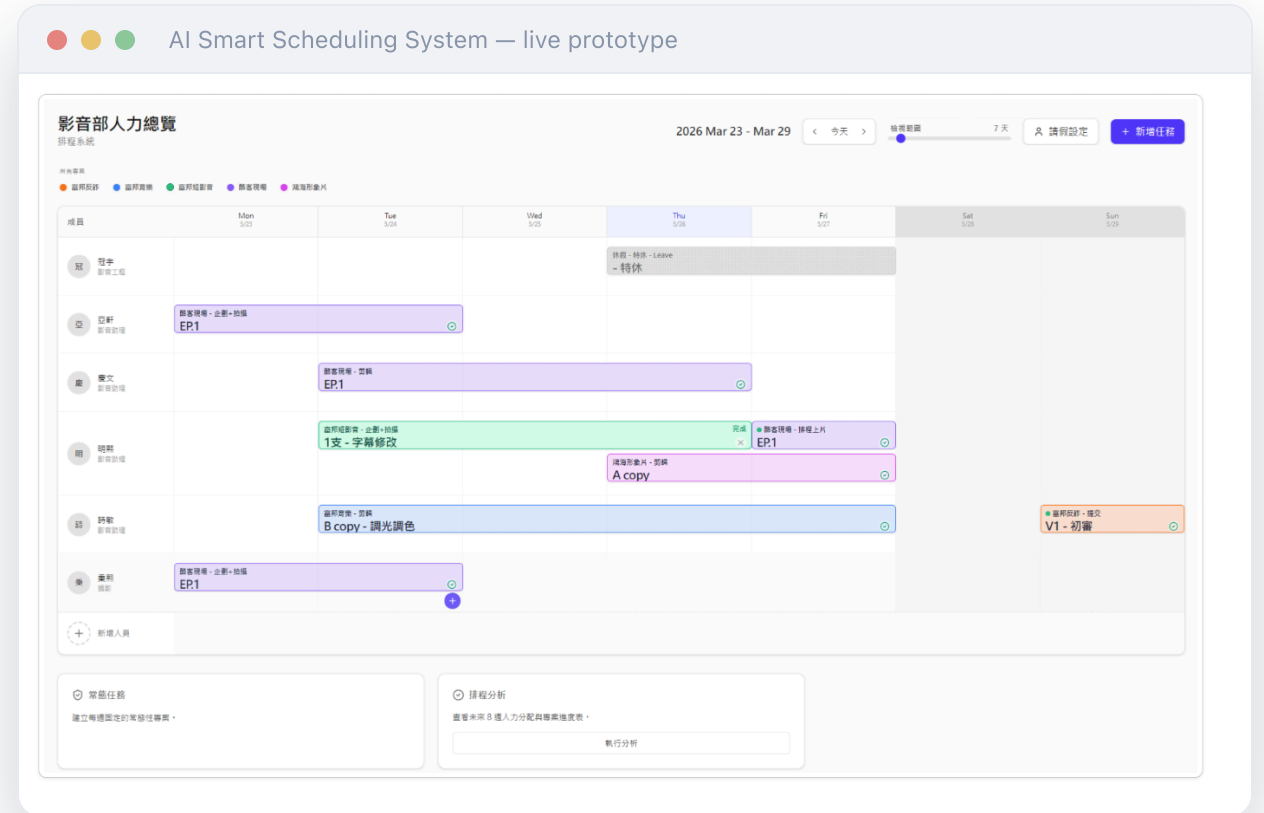
The manager **voluntarily rebuilt their own spreadsheets** to match the prototype — proof the design had already changed how they think.

Google AI Studio

6 core modules

Usability testing

Prototyped in 2 weeks



Turning this into a science-fair study

Ask a testable question

- Does an optimizing scheduler beat greedy or manual on conflicts?
- How does solve-time grow as the problem gets bigger?
- Which constraints matter most to a good schedule?

Design a fair test

- Define the metric & generate or collect real schedules
- Run each method on the same inputs
- Compare fairly, graph the trade-off, argue from it

Our role: we coach students through each step — framing the question, modelling the problem, building a prototype with AI tools, and presenting like a real researcher.

WHAT STUDENTS WALK AWAY WITH

Deliverables

01

A research question

A focused, testable hypothesis they can defend.

02

Data & analysis

Real measurements, a clear method, and honest graphs.

03

A working prototype

A real, runnable demo built with AI tools.

Poster & talk

Competition-ready board and a confident presentation.

Judge-ready Q&A

Practice fielding tough questions like a researcher.



AI SMART SCHEDULING · SAMPLE LESSON

From a messy spreadsheet to a research question

This is one of many topics your child can explore with us — real problems, real data, and a project worthy of the world stage.

World Science Academy

Helping Your Child Take the World Stage